Housing Affordability Dynamics

Price-to-Rent Ratios Across U.S. Metro Areas, January 2020–August 2026

Thao Nguyen

DATA 400 | Individual Mini-Project | Fall 2026

How did home values and rents change relative to one another and how did that relationship vary across metro areas?

Research question and motivation

Research question

How did the relationship between home values and rents evolve across major U.S. metropolitan areas from January 2020 to August 2026?

Why it matters

- Housing is a major household expense.
- Home values and rents can move at different rates.
- A changing price-to-rent ratio signals a shift in home values relative to rental costs.

What this study adds

- Tracks the national trend over time.
- Compares growth across metro areas.
- Tests whether rent growth is associated with later home-value growth.

Data and sample

Sources: Zillow Research

- **ZHVI**: typical home-value index.
- **ZORI**: observed rent index.
- Study period: January 2020–August 2026.
- Zillow metro identifiers used to align the datasets.

Source: `https://www.zillow.com/research/data/`

Sample construction

1. Select the 100 largest metro areas by Zillow's size ranking.
2. Require at least 76 of 80 expected monthly observations (95% coverage).
3. Keep months with both home-value and rent observations.

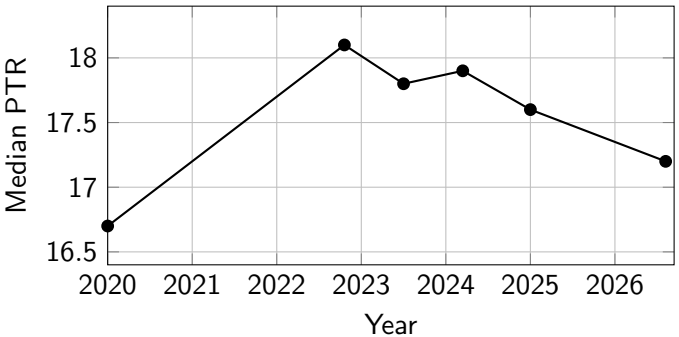
How the analysis works

Price-to-rent ratio (PTR)

$$\mathrm{PTR}_{i,t} = \frac{\mathrm{ZHVI}_{i,t}}{12 \times \mathrm{ZORI}_{i,t}}$$

- **PTR:** home value divided by annual rent. A higher ratio means buying is more expensive relative to renting.
- **Cumulative growth:** percentage change between January 2020 and August 2026.
- **Growth divergence:** home-value growth minus rent growth, in percentage points.
- **Lag analysis and regression:** test whether rent growth is associated with home-value growth up to 24 months later.
- **K-means clustering:** group metros by home-value and rent growth; standardize both measures and set $k = 3$.

Finding 1: PTR peaked in 2022, then eased



Key values

18.1 peak, late 2022

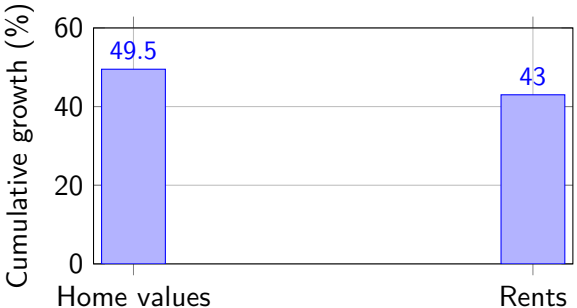
17.2 August 2026

16.7 January 2020

Illustrative trend based on the notebook's reported reference points; not a month-by-month reconstruction.

Interpretation: PTR rose through 2022, then eased but stayed above its 2020 level.

Finding 2: Home values grew faster overall



Typical metro

+5.8 pp median gap between home-value and rent growth

- Home values increased about 49.5%.
- Rents increased about 43.0%.
- Most metros saw home values grow faster, but the gap varied substantially.

Growth rates compare January 2020 with August 2026; the gap is in percentage points.

Finding 3: Metro-level outcomes differ

Largest growth divergence

McAllen, Texas

- Home values grew about 29 percentage points faster than rents.
- Rochester and Syracuse, New York, followed at roughly 22–23 points.

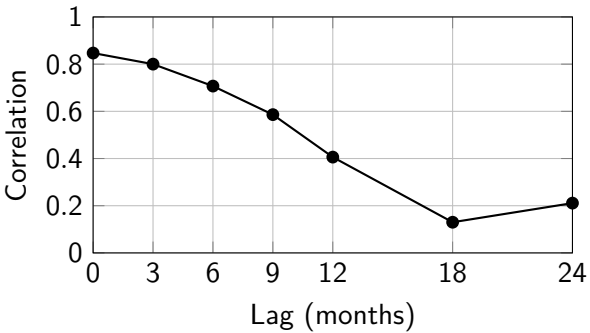
Largest PTR increase

San Jose, California

- PTR increased by about 3.4 points.
- Salt Lake City and McAllen followed.

Takeaway: A metro can experience a large growth gap or a large PTR increase; these are related but distinct measures.

Finding 4: Rent growth has limited predictive power



12-month regression

$$\hat{y} = 1.35 + 0.54x$$

$$R^2 = 0.165$$

- Positive slope: higher rent growth is associated with higher home-value growth one year later.
- $R^2 = 0.165$: the model explains 16.5% of variation.

Interpretation: Correlation weakens at longer lags; this is association, not causation.

Finding 5: Three broad growth profiles

Cluster profile	Home-value growth	Rent growth	Gap
Lower-growth markets	30.6%	28.1%	+2.5 pp
Moderate-growth markets	47.8%	40.3%	+7.5 pp
Higher-growth markets	58.3%	52.8%	+5.5 pp

- K-means groups metros using cumulative home-value and rent growth.
- The clusters distinguish lower-, moderate-, and higher-growth markets.
- These are descriptive groups, not official affordability categories.

Implications: who can use these findings?

Households

Compare how home values and rents have changed across metropolitan areas when considering where to live or whether to rent or buy.

Policymakers and planners

Use metro-level trends as a starting point for identifying markets that may warrant closer affordability monitoring.

Researchers and analysts

Compare housing-market trajectories across metros and use the results to motivate more detailed local analysis.

These indicators describe market trends; they do not determine affordability for any individual household.

Limitations and conclusion

Limitations

- Zillow indices summarize typical market trends, not every home or lease.
- The sample focuses on the 100 largest metros, so results may not generalize to smaller markets.
- PTR does not include household income, mortgage rates, taxes, insurance, maintenance, or transaction costs.
- Correlation and K-means clustering describe patterns; they do not establish causation.

Conclusion

From 2020 to August 2026, home values generally grew faster than rents. Median PTR peaked in 2022 and then declined, while remaining above its early-2020 level. Metro-level growth patterns varied substantially. Rent growth was positively associated with later home-value growth, but the regression had limited explanatory power ($R^2 = 0.165$).

Questions

Thank you

Questions and discussion